Estimating Power Zone Distributions in Cycling

From Gaussian Moment Matching to Bayesian Dirichlet Regression

Calum Moir

2026-05-12

Table of contents

Abstract

Cycling training data from platforms such as Garmin Connect and Strava typically reports only aggregate statistics per activity: average power, normalised power, and total duration. Without second-by-second power traces these summaries are insufficient to determine *how* training load was distributed across physiological zones, yet that distribution is critical for evaluating training balance and recommending future sessions. This paper describes two complementary statistical models implemented in a personal cycling performance dashboard. The first is an analytic Gaussian approximation that uses the relationship between normalised power and average power to infer the spread of the power distribution and thus allocate ride time across zones. The second is a Bayesian Dirichlet regression model trained on Garmin rides that carry ground-truth per-zone times, which learns how intensity and power variability predict zone distributions and generalises to rides that have only summary statistics. Together the models enable continuous, interpretable monitoring of training polarisation in line with evidence-based guidelines [5, 6].

1 Introduction

Training load monitoring in cycling has matured considerably over the past two decades. The *Performance Management Chart* (PMC), derived from Banister's impulse-response model [2], quantifies fitness (Chronic Training Load, CTL), fatigue (Acute Training Load, ATL) and form (Training Stress Balance, TSB) from a single daily metric — Training Stress Score (TSS). This framework, popularised by Coggan and Allen [3], allows athletes and coaches to plan and respond to training load systematically.

Alongside load management, *training intensity distribution* has attracted growing attention. Seiler and Kjerland [5] demonstrated that elite endurance athletes spend approximately 80% of training time at low intensity (below the first lactate threshold) and 15–20% at high intensity (near or above VO_2max), with very little time in the intermediate "grey zone" between the two thresholds. A randomised controlled trial by Stöggl and Sperlich [6] confirmed that this *polarised* distribution produced greater VO_2max gains than threshold or high-volume training models over a 9-week intervention.

Implementing such monitoring requires knowing not just *how much* training was done but *at what intensity*. When a cycling computer records second-by-second power, the distribution is fully observed; Garmin devices for instance store time-in-zone directly. But Strava activity exports frequently carry only summary statistics: average power, normalised power, and duration. The question this paper addresses is: *given only these summary statistics, can we estimate the zone time distribution with acceptable accuracy?*

The remainder of the paper proceeds as follows. Section 2 provides background on the relevant physiological and mathematical concepts. Section 3 formalises the estimation problem. Section 4 and Section 5 describe the two models. Section 6 summarises empirical results on the author's own training data. Section 7 discusses limitations and future directions.

Full mathematical derivations are collected in the appendix for readers who want to understand the models at a deeper level.

2 Background

2.1 The Performance Management Chart

The PMC tracks three exponentially weighted moving averages of daily TSS. Let s_t denote the TSS on day t. Fitness (CTL) and fatigue (ATL) evolve as:

$$\mathrm{CTL}_t = (1-\alpha_{\mathrm{CTL}})\,\mathrm{CTL}_{t-1} + \alpha_{\mathrm{CTL}}\,s_t$$

$$\mathrm{ATL}_t = (1-\alpha_{\mathrm{ATL}})\,\mathrm{ATL}_{t-1} + \alpha_{\mathrm{ATL}}\,s_t$$

where $\alpha_{\mathrm{CTL}} = 1-e^{-1/42}$ (42-day time constant) and $\alpha_{\mathrm{ATL}} = 1-e^{-1/7}$ (7-day time constant). Form, or Training Stress Balance, is $\mathrm{TSB}_t = \mathrm{CTL}_{t-1} - \mathrm{ATL}_{t-1}$.

The TSS for a ride lasting T seconds at normalised power NP watts is:

$$\mathrm{TSS} = \frac{T \cdot \mathrm{NP}^2}{\mathrm{FTP}^2 \cdot 3600} \times 100$$

where FTP is the athlete's functional threshold power.

2.2 Coggan Power Zones

Coggan [3] defined seven training zones as percentages of FTP:

Table 1: Coggan seven-zone power model.

Zone	Name	FTP fraction
Z1	Recovery	$< 55\%$
Z2	Endurance	55–75%
Z3	Tempo	76–90%
Z4	Threshold	91–105%
Z5	$\mathrm{VO_2max}$	106–120%
Z6	Anaerobic	121–150%
Z7	Neuromuscular	$> 150\%$

In the polarised framework, Z1+Z2 form the "easy" block, Z3+Z4 the "grey zone", and Z5–Z7 the "hard" block.

2.3 Normalised Power

Normalised power (NP) is the 4th-root of the mean 4th power of a rolling 30-second average of instantaneous power over the ride. Formally, for a sequence of power observations $\{x_t\}$:

$$\mathrm{NP} = \left(\frac{1}{n} \sum_{t=1}^{n} x_t^4 \right)^{1/4}$$

This is equivalent to the $\mathcal{L}^4$ norm of the empirical distribution, and it weights high-power efforts more heavily than average power. A ride with large spikes — such as a criterium or interval session — will have $\mathrm{NP} \gg \mathrm{AP}$, while a steady endurance ride will have $\mathrm{NP} \approx \mathrm{AP}$. The *variability index* $\mathrm{VI} = \mathrm{NP}/\mathrm{AP}$ encodes this spread concisely.

2.4 Seiler's Polarised Training Model

Seiler and Kjerland [5] observed that elite cross-country skiers, cyclists and rowers converge on a distribution in which roughly 75–80% of training sessions fall below the first ventilatory threshold and 15–20% above the second, irrespective of total volume. The grey zone between thresholds — which Coggan's Z3 and Z4 approximate — tends to be minimised.

This polarisation is thought to reflect an adaptation optimisation: easy sessions maximise fat oxidation and mitochondrial biogenesis without accumulating excess fatigue, while hard sessions push the VO_2max ceiling without the chronic inflammatory cost of sustained threshold work.

In practice the thresholds are athlete-specific. For power-based monitoring, Z3–Z4 (76–105% FTP) serves as a pragmatic proxy for the grey zone, with a flag raised when these zones exceed 15% of total training time.

3 The Zone Estimation Problem

Let a ride be characterised by a pair $(\mathrm{AP}, \mathrm{NP})$ and a total duration T seconds. Define FTP as the athlete's functional threshold power. The object of interest is the *zone time vector*:

$$\mathbf{t} = (t_1, t_2, t_3, t_4, t_5, t_6, t_7), \quad \sum_k t_k = T$$

where t_k is the seconds spent with power in zone k.

Equivalently, we work with the *zone fraction vector*:

$$\mathbf{z} = \mathbf{t}/T, \quad z_k \geq 0, \quad \sum_k z_k = 1$$

The problem: given only $(\text{AP}, \text{NP}, T)$ and the athlete's FTP, estimate $\mathbf{z}$.

Two approaches are developed:

1. **Gaussian moment matching** — treat instantaneous power as Gaussian, use AP and NP to determine the distribution parameters, and compute zone fractions as probabilities under this Gaussian.
2. **Bayesian Dirichlet regression** — use rides with observed $\mathbf{z}$ (from Garmin device data) to fit a regression model from (AP, NP) to $\mathbf{z}$, exploiting the natural compositional structure of zone fractions.

4 Gaussian Moment Matching

4.1 Intuition

For a steady endurance ride at 65% FTP, power barely deviates from the mean: almost all time is spent in Z2. For a mixed interval session at the same *average* power but with repeated spikes into Z5–Z6, the power distribution is much wider. The key insight is that NP encodes this spread through the 4th moment.

4.2 The Model

Assume instantaneous power X follows a normal distribution:

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

Set $\mu = \text{AP}$ (average power equals the mean of the distribution). The 4th raw moment of a normal distribution is:

$$\mathbb{E}[X^4] = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$$

Since $\text{NP} = (\mathbb{E}[X^4])^{1/4}$, we have $\mathbb{E}[X^4] = \text{NP}^4$. Substituting and collecting in $s = \sigma^2$:

$$3s^2 + 6\mu^2 s - (\text{NP}^4 - \mu^4) = 0$$

Solving the positive root:

$$s^* = \frac{-3\mu^2 + \sqrt{6\mu^4 + 3 \cdot \mathrm{NP}^4}}{3}, \qquad \sigma = \sqrt{\max(s^*, 0)}$$

The derivation is given in full in Section 9.2.

4.3 Zone Fractions

With μ and σ determined, the fraction of time in zone k (bounded by FTP fractions $[\ell_k, u_k)$) is:

$$z_k = P(\ell_k \cdot \mathrm{FTP} \leq X < u_k \cdot \mathrm{FTP}) = \Phi\left(\frac{u_k \cdot \mathrm{FTP} - \mu}{\sigma}\right) - \Phi\left(\frac{\ell_k \cdot \mathrm{FTP} - \mu}{\sigma}\right)$$

where Φ is the standard normal CDF, computed via the error function $\Phi(x) = \frac{1}{2}[1 + \mathrm{erf}(x/\sqrt{2})]$.

4.4 Behaviour

- **Steady ride** (VI $\approx$ 1.02): σ is small, almost all probability mass sits in the zone containing AP, matching physical intuition.
- **Interval session** (VI $\approx$ 1.15–1.25): σ grows substantially, spreading time across Z1 through Z5+.
- **Limitation**: the model is unimodal. A true interval session has a *bimodal* distribution — easy recovery valleys and hard work peaks — not a single Gaussian. This leads to a slight underestimation of time at the extremes and overestimation near the mean. The Bayesian model addresses this by learning the actual distribution from data.

4.5 Python Implementation

```python
from math import erf, sqrt

def gaussian_zone_times(avg_power, np_power, duration_s, ftp, zones):
    """Estimate zone seconds using Gaussian moment matching."""
    mu    = avg_power
    inner = 6.0 * mu**4 + 3.0 * np_power**4
    s     = (-3.0 * mu**2 + sqrt(inner)) / 3.0
    sigma = sqrt(max(0.0, s))
```

```
    def cdf(x):
        if sigma <= 0:
            return 1.0 if x >= mu else 0.0
        return 0.5 * (1.0 + erf((x - mu) / (sigma * sqrt(2.0))))

    result = {}
    for zone_name, lo_frac, hi_frac in zones:
        hi_cdf = 1.0 if hi_frac >= 9.0 else cdf(hi_frac * ftp)
        result[zone_name] = duration_s * max(0.0, hi_cdf - cdf(lo_frac * ftp))
    return result
```

5 Bayesian Dirichlet Regression

5.1 Motivation

The Gaussian model is fully analytic and requires no data. Its limitation is the unimodal assumption. For riders with Garmin computers, *ground-truth* zone data is available: the device records actual seconds in each zone from the raw power stream. This section describes a model that learns from this data.

5.2 The Dirichlet Distribution

Zone fractions live in the probability simplex $\Delta^6 = \{\mathbf{z} \in \mathbb{R}^7 : z_k \geq 0, \sum_k z_k = 1\}$. The Dirichlet distribution is the natural distribution over this space:

$$\mathbf{z} \sim \mathrm{Dir}(\alpha), \quad \alpha_k > 0$$

with density:

$$p(\mathbf{z} \mid \alpha) = \frac{\Gamma(\alpha_0)}{\prod_k \Gamma(\alpha_k)} \prod_k z_k^{\alpha_k - 1}, \quad \alpha_0 = \sum_k \alpha_k$$

The mean is $\mathbb{E}[z_k] = \alpha_k / \alpha_0$, and the *precision* α_0 controls how concentrated the distribution is around its mean. Full properties are given in Section 9.3.

5.3 The Regression Model

For ride i with features intensity $q_i = \mathrm{AP}_i/\mathrm{FTP}$ and log variability index $v_i = \log(\mathrm{VI}_i)$, the concentration parameters are modelled as:

$$\log \alpha_{ik} = \beta_{0k} + \beta_{1k}\, q_i + \beta_{2k}\, v_i, \quad k = 1, \dots, 7$$

This log-linear link ensures $\alpha_{ik} > 0$ for all inputs. The full model is:

$$\begin{aligned} \beta_{jk} &\sim \mathcal{N}(0,1), \quad j \in \{0,1,2\},\ k = 1, \dots, 7 \\ \log \alpha_{ik} &= \beta_{0k} + \beta_{1k}\, q_i + \beta_{2k}\, v_i \\ \mathbf{z}_i &\sim \mathrm{Dir}(\alpha_i) \end{aligned}$$

The prior $\mathcal{N}(0,1)$ on log-scale is weakly informative — it allows α_k to vary between roughly 0.05 and 20, covering the full range from near-uniform to highly concentrated zone distributions.

5.4 Feature Engineering

The two features were chosen to be interpretable and sufficient:

Intensity $q_i = \mathrm{AP}_i/\mathrm{FTP}$ locates the average effort on the zone scale. A ride at $q = 0.65$ is deep in Z2; at $q = 0.92$ it sits at the threshold of Z4.

Log variability index $v_i = \log(\mathrm{VI}_i)$ where $\mathrm{VI} = \mathrm{NP}/\mathrm{AP}$ encodes power distribution spread. The log transform is natural: $\mathrm{VI} \geq 1$ always, so $v_i \geq 0$, and the log scale compresses the range for highly variable rides. Importantly, this feature connects back to the Gaussian model: VI is a monotone function of σ via the 4th moment equation, so both models share the same physical interpretation of power variability.

5.5 Posterior Inference

Posterior sampling uses the No-U-Turn Sampler (NUTS) [4] implemented in PyMC [1]. Two chains of 1000 draws each (after 500 tuning steps) are run with target acceptance probability 0.90. Posterior means $\hat{\beta}_{jk}$ are stored for fast prediction; full posterior samples are optionally retained for uncertainty quantification.

5.6 Prediction

For a new ride with $(\mathrm{AP}, \mathrm{NP})$:

$$\hat{\alpha}_k = \exp\Big(\hat{\beta}_{0k} + \hat{\beta}_{1k}\,(\mathrm{AP}/\mathrm{FTP}) + \hat{\beta}_{2k}\,\log(\mathrm{NP}/\mathrm{AP})\Big)$$

$$\hat{z}_k = \frac{\hat{\alpha}_k}{\sum_j \hat{\alpha}_j}$$

When full posterior samples are available, uncertainty in $\hat{\mathbf{z}}$ can be propagated by drawing $\alpha^{(s)}$ from each posterior sample s and reporting posterior predictive means and credible intervals.

5.7 Python Implementation

```
import pymc as pm
import pytensor.tensor as pt

def build_and_fit(intensity, log_vi, zone_fracs):
    """
    intensity  : array [N] - AP/FTP per ride
    log_vi     : array [N] - log(NP/AP) per ride
    zone_fracs : array [N, 7] - observed zone fractions (Garmin ground truth)
    """
    N_ZONES = 7
    with pm.Model():
        beta0 = pm.Normal("beta0", mu=0.0, sigma=1.0, shape=N_ZONES)
        beta1 = pm.Normal("beta1", mu=0.0, sigma=1.0, shape=N_ZONES)
        beta2 = pm.Normal("beta2", mu=0.0, sigma=1.0, shape=N_ZONES)

        intensity_t = pt.as_tensor_variable(intensity[:, None])   # [N, 1]
        log_vi_t    = pt.as_tensor_variable(log_vi[:, None])      # [N, 1]

        log_alpha = beta0 + beta1 * intensity_t + beta2 * log_vi_t  # [N, 7]
        alpha = pm.math.exp(log_alpha)

        pm.Dirichlet("obs", a=alpha, observed=zone_fracs)
```

```
    return pm.sample(draws=1000, tune=500, target_accept=0.90)
```

5.8 Method Priority in the Dashboard

The dashboard applies a three-level fallback per ride:

1. **Garmin device data** (`power_z1..power_z7`): exact seconds recorded on-device from raw power. Used whenever non-zero zone values are present.
2. **Bayesian prediction**: applied when the model is fitted and Garmin data is absent (typical for Strava-only rides).
3. **Gaussian fallback**: used when neither Garmin data nor a fitted model is available.

This hierarchy ensures the most accurate available estimate is always used, while remaining gracefully degradable.

6 Results

The dashboard was evaluated on the author's training data: 25 cycling activities from the past four weeks comprising Garmin-recorded rides and Strava imports from Zwift and outdoor rides.

The naive approach — assigning each ride entirely to the zone containing its average power — produced a nonsensical distribution: 98.8% Z1/Z2, 0% in any zone above Z2, since most endurance rides average into Z2 and the entire duration was assigned there.

After applying the Gaussian model, the four-week distribution became:

Table 2: Four-week zone distribution from Gaussian model.

Zone	% of total time
Z1 Recovery	32.0%
Z2 Endurance	22.7%
Z3 Tempo	24.9%
Z4 Threshold	11.0%
Z5 VO_2max	4.6%
Z6 Anaerobic	3.6%
Z7 Neuromuscular	1.2%

The grey-zone total (Z3+Z4 = 35.9%) substantially exceeded the 15% target, triggering the polarised audit flag. The recommendation engine correctly suggested reducing tempo work and adding a VO_2max session, in alignment with Seiler's guidelines.

The Bayesian Dirichlet model is trained on Garmin rides with per-zone data. Fitted posterior means for β_1 (intensity coefficient) confirm that higher average power relative to FTP concentrates time in higher zones, as expected. The β_2 (log-VI coefficient) is positive for Z5–Z7 and negative for Z1–Z2, correctly encoding that more variable rides (higher VI) shift time toward the extremes.

7 Discussion

7.1 Gaussian Model Strengths and Limitations

The analytic Gaussian approach requires no training data and performs well on rides with moderate variability. Its main limitation is the unimodal assumption: polarised interval sessions with long easy recoveries and hard work intervals are genuinely bimodal, and the Gaussian model will overestimate grey-zone time while underestimating Z1 and Z5+. For such sessions, VI often exceeds 1.20, and users should interpret zone estimates as approximations.

7.2 Bayesian Model Strengths and Limitations

The Dirichlet regression can learn more complex, data-driven mappings and is not constrained to a unimodal power distribution. Its limitation is the need for sufficient Garmin training data — the model requires at least five rides with non-zero per-zone data to fit. A cold-start problem exists for new athletes or those without a Garmin device: the Gaussian fallback applies until enough data accumulates.

The choice of $\mathcal{N}(0,1)$ priors on log-α is weakly informative. In a fully Bayesian setting, one could encode knowledge that endurance athletes typically spend most time in Z1/Z2 by placing informative priors on β_0. This is left as a future extension.

7.3 Dashboard Integration

Both models integrate directly with the polarised training audit and the workout recommendation engine. The audit computes:

- $p_{\text{easy}} = z_1 + z_2$ (target: $\geq 80\%$)
- $p_{\text{grey}} = z_3 + z_4$ (target: $\leq 15\%$)

- $p_{\text{hard}} = z_5 + z_6 + z_7$ (target: $\geq 15\%$)

The recommender steers workout selection away from grey-zone sessions when $p_{\text{grey}} > 15\%$ and toward VO_2max intervals when $p_{\text{hard}} < 5\%$.

7.4 Future Directions

Several natural extensions exist:

- **Bimodal model**: a mixture of two Gaussians (easy and hard) would better represent true interval sessions.
- **Ride-type conditioning**: including a categorical feature for ride type (outdoor, Zwift, group ride) could improve prediction accuracy.
- **Online learning**: updating the posterior incrementally as new Garmin rides arrive, rather than refitting from scratch.
- **Garmin workout push**: using the `garth` library to push structured workouts (generated by the recommender) to Garmin Connect and sync to the device.

8 Conclusion

This paper presented two statistical approaches to estimating cycling power zone distributions from summary statistics alone. The Gaussian moment-matching model provides an interpretable, parameter-free baseline grounded in the mathematical relationship between normalised power and the 4th moment. The Bayesian Dirichlet regression model extends this by learning from Garmin ground-truth data, with full uncertainty quantification via posterior sampling. Together they enable meaningful training polarisation monitoring even when only summary data is available — a practical advance for athletes training across multiple platforms.

Both models are implemented in an open-source cycling dashboard and form the basis of an automated recommendation engine that steers training away from the grey zone and toward evidence-based polarised load distributions.

Acknowledgements

The models, system architecture, and analytical direction described in this paper were designed by the author. The code implementation, mathematical write-up, and document were produced with

the assistance of Claude (Anthropic), an AI assistant, working from the author's specifications and iterative feedback throughout development.

---

9 Mathematical Appendix

9.1 Normalised Power as an $\mathcal{L}^4$ Norm

Normalised power is defined as the 4th root of the mean 4th power of a smoothed power signal. For a stationary power process $\{X_t\}$ with empirical distribution F:

$$\mathrm{NP} = \left(\int x^4 \, dF(x) \right)^{1/4} = \|X\|_4$$

This is the $\mathcal{L}^4$ norm of the power distribution, i.e. the 4th raw moment raised to the power 1/4.

The 4th power weighting is physiologically motivated: the metabolic stress of cycling at high power is disproportionately greater than at low power (due to the exponential relationship between power output and glycolytic demand near $\mathrm{VO_2max}$). A ride with repeated near-maximal efforts is therefore more stressful than a steady ride at the same average power, and NP captures this nonlinearly.

Why not $\mathcal{L}^2$ (RMS)? The RMS ($\|X\|_2$) would equal $\sqrt{\mu^2 + \sigma^2}$ under a Gaussian, giving only a small boost for variable rides. The $\mathcal{L}^4$ norm amplifies the effect of high-power spikes much more aggressively, matching empirical observation that interval sessions produce disproportionate fatigue.

9.2 Derivation of Gaussian σ from NP and AP

Setup. Assume $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\mu = \mathrm{AP}$.

Fourth raw moment. For a normal distribution:

$$\mathbb{E}[X^4] = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$$

This follows from the moment generating function or Isserlis's theorem applied to the normal distribution.

Setting the 4th moment equal to NP^4:

$$\mathrm{NP}^4 = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$$

Rearranging as a quadratic in $s = \sigma^2$:

$$3s^2 + 6\mu^2 s - (\mathrm{NP}^4 - \mu^4) = 0$$

Applying the quadratic formula with $a = 3$, $b = 6\mu^2$, $c = -(\mathrm{NP}^4 - \mu^4)$:

$$s = \frac{-6\mu^2 + \sqrt{36\mu^4 + 12(\mathrm{NP}^4 - \mu^4)}}{6} = \frac{-3\mu^2 + \sqrt{6\mu^4 + 3 \cdot \mathrm{NP}^4}}{3}$$

(The negative root is discarded as $s = \sigma^2 \geq 0$.)

Therefore:

$$\boxed{\sigma = \sqrt{\max\left(0,\ \frac{-3\mu^2 + \sqrt{6\mu^4 + 3 \cdot \mathrm{NP}^4}}{3}\right)}}$$

Limiting cases:

- NP = AP (VI = 1): substituting $\mathrm{NP} = \mu$ gives $s = 0$, so $\sigma = 0$. All power is at exactly AP — physically, a perfectly constant power output.
- NP $\gg$ AP: σ grows, spreading the distribution widely across zones.

Note on stationarity. The derivation treats $\{X_t\}$ as an i.i.d. sample from $\mathcal{N}(\mu, \sigma^2)$. Real rides are not i.i.d. — power varies continuously and exhibits autocorrelation. The derivation nonetheless provides a reasonable approximation because the 4th moment is a global property of the power distribution that does not depend on its temporal structure.

9.3 Properties of the Dirichlet Distribution

The Dirichlet distribution with parameter vector $\alpha = (\alpha_1, \ldots, \alpha_K)$, $\alpha_k > 0$, is the conjugate prior for the Categorical distribution. Its probability density function on the simplex Δ^{K-1} is:

$$p(\mathbf{z} \mid \alpha) = \frac{\Gamma(\alpha_0)}{\prod_{k=1}^{K} \Gamma(\alpha_k)} \prod_{k=1}^{K} z_k^{\alpha_k - 1}$$

where $\alpha_0 = \sum_k \alpha_k$ is the *concentration* (or *precision*).

Moments:

$$\mathbb{E}[z_k] = \frac{\alpha_k}{\alpha_0}, \qquad \mathrm{Var}[z_k] = \frac{\alpha_k(\alpha_0 - \alpha_k)}{\alpha_0^2(\alpha_0 + 1)}$$

The variance is small when α_0 is large (concentrated distribution) and approaches the maximum when $\alpha_0 \to 0$ (distribution spreading toward the vertices of the simplex).

Marginals. Each marginal $z_k \sim \mathrm{Beta}(\alpha_k, \alpha_0 - \alpha_k)$, so the expected fraction in zone k is exactly α_k/α_0.

Interpretation. The Dirichlet is the natural model for *compositional data* — vectors of non-negative fractions summing to 1. Zone time fractions are exactly this type of data. Modelling them with a multivariate normal would ignore the sum-to-one constraint and risk producing negative fractions or fractions exceeding 1.

9.4 Full Dirichlet Regression Model Specification

9.4.1 Notation

Symbol	Description
N	Number of training rides
$K = 7$	Number of Coggan power zones
q_i	Intensity feature: $\mathrm{AP}_i/\mathrm{FTP}$
v_i	Variability feature: $\log(\mathrm{VI}_i) = \log(\mathrm{NP}_i/\mathrm{AP}_i)$
$\mathbf{z}_i$	Observed zone fraction vector (from Garmin), length K
α_i	Concentration parameter vector for ride i
β_{jk}	Regression coefficient for feature j, zone k

9.4.2 Likelihood

$$\mathbf{z}_i \mid \alpha_i \sim \mathrm{Dir}(\alpha_i)$$

$$\log \alpha_{ik} = \beta_{0k} + \beta_{1k}\, q_i + \beta_{2k}\, v_i$$

9.4.3 Priors

$$\beta_{jk} \sim \mathcal{N}(0, 1), \quad j \in \{0, 1, 2\},\ k \in \{1, \dots, 7\}$$

Total parameters: $3 \times 7 = 21$ scalar coefficients.

9.4.4 Prior Predictive Check

Under the prior, when $q = 0.65$ (a Z2 ride, $\beta_{jk} \sim \mathcal{N}(0, 1)$):

$$\log \alpha_k \sim \mathcal{N}(0, 1 + 0.65^2 + 0^2) = \mathcal{N}(0, 1.42)$$

So $\alpha_k \sim \text{LogNormal}(0, 1.42)$, which has a median of 1 and a 95% interval of roughly $(0.04, 23)$. This is weakly informative: it allows anything from a near-uniform distribution over zones to one highly concentrated in a single zone, without strongly constraining either. The prior is appropriate given the wide variability in athlete training patterns.

9.4.5 Posterior Inference

The joint posterior:

$$p(\beta \mid \mathbf{z}_{1:N}) \propto p(\beta) \prod_{i=1}^{N} p(\mathbf{z}_i \mid \alpha_i(\beta))$$

has no closed form. NUTS [4] is used to draw samples. The $\mathcal{L}^4$ Riemannian geometry of the Dirichlet likelihood makes the posterior mildly non-Euclidean; the `target_accept=0.90` setting in PyMC ensures the sampler takes smaller steps, reducing divergences near the boundary of the simplex.

9.4.6 Posterior Predictive Distribution

For a new ride $*$ with features (q_*, v_*):

$$\hat{\alpha}_{*k}^{(s)} = \exp\Big(\beta_{0k}^{(s)} + \beta_{1k}^{(s)}\, q_* + \beta_{2k}^{(s)}\, v_*\Big)$$

$$\hat{z}_{*k}^{(s)} = \frac{\hat{\alpha}_{*k}^{(s)}}{\sum_j \hat{\alpha}_{*j}^{(s)}}$$

for each posterior sample $s = 1, \ldots, S$. The posterior predictive mean is:

$$\hat{z}_{*k} = \frac{1}{S} \sum_{s=1}^{S} \hat{z}_{*k}^{(s)}$$

and 94% credible intervals follow from the empirical 3rd and 97th percentiles of $\{\hat{z}_{*k}^{(s)}\}$.

9.5 Identifiability Note

The regression model for $\log \alpha$ is identifiable because the features q_i and v_i take different values across rides, and the Dirichlet likelihood distinguishes between different α vectors (unlike the Categorical likelihood, which is invariant to rescaling of α). In practice the posterior is well-concentrated with 30+ training rides.

References

[1] Oriol Abril-Pla et al. “PyMC: A Modern and Comprehensive Probabilistic Programming Framework in Python”. In: *PeerJ Computer Science* 9 (2023), e1516. DOI: 10.7717/peerj-cs.1516.

[2] Eric W. Banister. “Modeling elite athletic performance”. In H.J. Green, J.D. McDougal, and H. Wenger (Eds.), Physiological Testing of Elite Athletes, Human Kinetics, Champaign IL. 1975.

[3] Andrew R. Coggan and Hunter Allen. *Training and Racing with a Power Meter*. 2nd. Boulder, CO: VeloPress, 2010.

[4] Matthew D. Hoffman and Andrew Gelman. “The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo”. In: *Journal of Machine Learning Research* 15 (2014), pp. 1593–1623.

[5] K. Stephen Seiler and Glenn Øyvind Kjerland. “Quantifying training intensity distribution in elite endurance athletes: is there evidence for an “optimal” distribution?” In: *Scandinavian Journal of Medicine & Science in Sports* 16.1 (2006), pp. 49–56. DOI: 10.1111/j.1600-0838.2004.00418.x.

[6] Thomas Stöggl and Billy Sperlich. “Polarized training has greater impact on key endurance variables than threshold, high intensity, or high volume training”. In: *Frontiers in Physiology* 5 (2014), p. 33. DOI: 10.3389/fphys.2014.00033.